

Shadab Raza

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EDUCATION

Sitare University

Bachelor of Technology in Computer Science

Lucknow, India

Oct. 2022 – May 2026

EXPERIENCE

FirstHive

Software Engineer

Bengaluru, Karnataka

June 2025 – Present

- Developed and maintained full-stack web applications using React.js, TypeScript, Redux, FastAPI, Flask, and MySQL, delivering analytics dashboards.
- Built and optimized REST APIs with filtering, pagination, and SQL aggregations, improving reporting capabilities and backend performance.
- Designed scalable frontend and backend architectures by separating business logic, API services, and UI components, enhancing maintainability and code reusability.
- Improved application performance through MySQL query optimization, React memoization techniques (**useMemo**, **React.memo**), and efficient state management.
- Built responsive user interfaces and data visualization features, including hierarchical reports, CSV bulk uploads, validation, and error-handling mechanisms.

Zeni.ai

Quality Assurance Intern

Pune, Maharashtra

May 2024 – Aug. 2024

- Maintained and optimized Pytest-based automated test suites, including comprehensive smoke and regression test cases, ensuring stable, reliable, and high-quality software releases across multiple deployments.

Skyroot Aerospace

SDE Summer Intern

Hyderabad, Telangana

June 2023 – Aug. 2023

- Developed Python scripts using NumPy and Pandas to analyze structured dataframe data, extracting key parameters such as pressure, temperature, and speed across different stages of the motor engine, enabling data-driven insights and performance evaluation.

PROJECTS

Optimal City Routing System | Python, Graph Theory, A*, GBFS, BFS, DFS

[🔗 Source Code](#)

- Developed an intelligent city routing system using **A***, **GBFS**, **BFS**, and **DFS** algorithms for optimal pathfinding across a randomly connected graph.
- Implemented heuristic pathfinding techniques and analyzed **routing** performance using execution time, memory usage, and path optimality.

CodeHub – Online Judge | React, Node.js, Express, MongoDB, Docker, AWS, Gemini AI

[🔗 Source Code](#)

- Developed **CodeHub**, an online judge platform delivering verdicts, a leaderboard, and AI code suggestions.
- Achieved secured code execution via **Docker** containerisation; supports 4 languages (C++, C, Java, Python).
- Designed and implemented tags, difficulty levels, dashboard, and a custom compiler using Node.js child process.
- Deployed on **AWS** using **ECR** & **EC2**.

MyHealthPal – AI Mental Health Platform | React, Node.js, FastAPI, BERT, Python

[🔗 Source Code](#)

- Built a dual AI mental health platform with **therapy (BERT)** and diagnosis (Claude Sonnet via OpenRouter) sessions.
- Fine-tuned a **BERT** model using Hugging Face Transformers for real-time sentiment classification; improved relevance using **TF-IDF** and cosine similarity.
- Developed a voice-enabled UI with speech-to-text, TTS, and animated chatbot, plus geolocation-based healthcare provider discovery.

TECHNICAL SKILLS

AI/ML: Machine Learning, LangChain, CrewAI, HuggingFace, SentenceTransformers, PyTorch, TensorFlow, Keras, Prompt Engineering, LLMs, RAG, AI Agents

Languages & Frameworks: Python, Flask, FastAPI, SQLAlchemy, Express.js, Node.js, React.js, TypeScript, Redux, JavaScript (ES6+), HTML5, CSS3

Databases: MySQL, MongoDB, SQLite, Supabase

Developer Tools: Git, GitHub, GitLab, Postman, Jira, Claude Code, Cursor AI

Build Tools: NPM, Vite, Pip, Webpack, Babel